INCH-POUND

MIL-P-70695 (AR)
AMENDMENT 3
29 September 1994
SUPERSEDING
AMENDMENT 2
31 August 1993

MILITARY SPECIFICATION

PROJECTILE, PRACTICE, 155MM, M804E1
CAST METAL PARTS ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-70695 (AR), dated 15 November 1991, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Change "M804E1" to "M804A1"

* 1.1 Line 2: Change "M804E1" to M804A1"

PAGE 2

2.1.2 DRAWINGS, Delete the following drawings in their entirety:

"7548993, 12913926, 12913928 and 12913929"

* 2.1.2 DRAWINGS, 12913927: Change: "M804E1" to M804A1"

2.1.2 DRAWINGS, Add:

"INSPECTION EQUIPMENT DRAWINGS

8593429 - Band Seating Testing Machine; Indentor and Test Data
9280429 - Rotating Band Seat Tester"

2.1.2 PUBLICATIONS: Delete in its entirety (Title and Text)

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PAGE 3

2.2 Add the following ASTM:

"ASTM E604 - Dynamic Tear Testing of Metallic Materials"

3.2, Line 3, Delete "Drawing 12913928" and substitute "Drawing 12913927".

PAGE 5

* 3.6.2 Line 3: Delete the words "applicable specifications" and substitute "applicable drawings and specification requirements"

3.7.2 Line 2, Insert the following parenthetical line between the words "defect" and "will":

"(not specifically covered in 3.7.3.b)"

PAGE 6

3.7.3.b.1: Delete line 1 in its entirety and substitute the following:

"All indications in the following categories shall be better than Level I;"

3.7.3.b.2 Delete this sub-paragraph in its entirety.

3.7.3.b.3 Line 2 Delete, "No worse than level III, of category cc," and substitute the following:

"No worse than level III, of category CA, CB, CC or CD,"

PAGE 8

3.10.1 Line 1, Delete "Preeparation" and substitute "Preparation".

PAGE 10

4.3.2.1.2 Line 3; Delete "4.4.3.1" and substitute 4.3.2.1.1".

Line 4; Delete "3.10" and substitute "3.11".

PAGES 11 & 12

* Delete "TABLE I" and substitute the following included "TABLE I"

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PAGE 15

4.4.2.1 Sheet 1 of 2:

Drawing Number Block: Delete "12913929/12913928" and substitute "12913929".
Major 103; Delete "Depth of scoring and substitute "Depth of tool marks".
Major 104; Delete "Scoring or Knurling missing from rotating band seat" and substitute "Knurling missing from rotating band seat".
Major 117: Delete "4.4.3.5" and substitute "4.4.3.6".

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	1 OF 2	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Blank Rotating Band			
	(Dwg. 7548993)			
	Mechanical properties	5, 100%	3.6.2	4.5.3.2.1.1
	Body (Unpainted)			
	(Dwg. 12913929)			
	Examination for defects 1/	5, 100%	3.1	4.4.2.1
	Hydrostatic 2/	5, 100%	3.7.3a	4.5.6
	Radiographic 2/	5, 100%	3.7.3b	4.5.5
	Mechanical properties 2/	5, 100%	3.5.1	4.5.2
	Microstructure properties 2/	5, 100%	3.9	4.5.7
	Body Assembly			
	(Dwg 12913928)			
	Examination for defects 1/	5, 100%	3.1	4.4.2.2
	Band seating 3/	5, 100%	3.6	4.5.3.1
	Mechanical properties of rotating band 3/	5, 100%	3.5.2	4.5.3.2.1
	Sidewall contact of rotating band 3/	5, 100%	3.6.3	4.5.3.2.2

NOTES:
1. The examination for defects are only for dimensional and visual inspections in paragraph 4.4.2.
2. After the dimensional and visual inspection of the 5 samples is completed, subject these samples to hydrostatic test, radiographic inspection then to the destructive tests of mechanical properties and microstructure.
3. After the dimensional and visual inspections of the 5 samples is completed, subject these samples to band seating, mechanical properties of rotating band and sidewall contact of rotating band.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile Assembly and Components	2 OF 2	See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Projectile Metal Parts Assembly			
	(Dwg. 12913927)			
	Examination for defects 1/	45, 100%	3.2	4.4.2.3
	Protective coating thickness 4/	45, 100%	3.10.2	4.5.4.1
	Salt spray testing 4/	5, 100%	3.2	4.5.4.2
	Functional test (Proving Ground)			
	(Dwg. 12913927)			
	Ballistic testing	10, 100%	3.11	4.5.8

NOTES:
1. The examination for defects are only for dimensional and visual inspections in paragraph 4.4.2.
4. After the dimensional and visual inspections of the 45 samples is completed, subject these same 45 samples to protective coating thickness, followed by salt spray testing of 5 of these samples.

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4.2.1 Sheet 2 of 2:

Drawing Number Block: Delete "12913429/12913928" and substitute "12913929"
Minor 216: Delete "Length from rear face to base diameter" and substitute "Length from rear face to basic diameter".

PAGE 17

4.4.2.2 Sheet 1 of 2:

Minor 202: Delete "Marking missing" and substitute "Marking (stamping) missing".

PAGE 18

4.4.2.2 Sheet 2 of 2:

Minor 214, Delete in its entirety.

PAGE 19

* 4.4.2.3 Title block, Delete "M804" and substitute "M804A1".

* 4.4.2.3 Add major 104:

"104", salt spray test, "4.4.3.5.1", "3.2", "4.5.4.2"

PAGE 20

4.4.2.4 Drawing Number block:

Delete "9331807/9334433" and substitute "12953254".

PAGE 23

4.5.2.1 a., line 2: delete "grinding to a sufficient depth" and substitute "grinding a flat to a sufficient depth"

PAGE 24

4.5.3.1.1 Line 2, 3 and 4: delete "prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 2.1) for test of 155MM, Proj., M107." and substitute "described in Drawing No. 9280429."

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4.5.3.2 Delete in its entirety and substitute the following:

"4.5.3.2 Mechanical properties and sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact."

PAGE 25

Figure 2, Delete the Title "Dynamic tear test bars." and substitute the following:

"Dynamic tear test bars. Approximate Location of Tension and Dynamic tear test bar specimens (T-Tensile, D-Dynamic tear)"

Add a note toward the right of figure and just above the horizontal centerline.

"Centerline of rotating band seat"

PAGE 26

* 4.5.3.2.1 Title: Delete title "mechanical properties" and substitute:

"Mechanical properties of disassembled rotating band"

* Add new paragraph 4.5.3.2.1.1:

4.5.3.2.1.1 Mechanical properties of annealed rotating band blanks. The band blank shall be cut from front to rear and flattened, without hammering which may further cold work the material, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a specimen with a 0.500 inch gage width. In any event, the gage cross section shall be determined to the nearest 0.001 square inch and utilized to calculate the tensile and yield strengths, and elongation.

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4.5.6 Hydrostatic test, line 4: Delete "3.6.3a" and substitute "3.7.3a".

4.5.6 Microstructure examination. Delete paragraph No. "4.5.6 and title and substitute the following: "4.5.7 Microstructure examination."

PAGE 29

* 6.7 Line 5, Delete "SMCAR-QAR-Q and SMCAR-FSA-SS" and substitute "SMCAR-QAR-R, SMCAR-FSA-MS and SMCAR-CCH-P"

6.7 Line 7: delete "61299."and substitute "61299-6000."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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